Graph Neural Networks for Functional Connectivity Analysis: Uncovering Neural Network Disruptions in Parkinson's Disease

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Abstract:
In the field of Parkinson's' disease diagnosis and progression monitoring, functional and structural connection patterns in the brain, which can be observed by modalities such as functional magnetic resonance imaging (MRI) and electroencephalography (EEG) are becoming more recognized as promising bio markers. An innovative use of Graph Neural Network (GNNs) is proposed in this study for analyzing patterns of brain connectivity. By utilizing the graph structure of brain networks, the researchers identify disturbances that are unique to Parkinson's disease conditions. In the proposed framework, brain regions are represented as nodes, and the edges of the graph correspond to functional or structural connections derived from either fMRI or EEG data, depending on the modality used for analysis. The investigation of the brain regions and the connection that are most pertinent to the diagnosis can be accomplished through the utilization of attention weights in feature importance analysis. In recognizing brain network changes in Parkinson's disease, the findings indicates that the GNN perform much better than traditional machine learning methods. Additionally, these research offers significant advancements into mechanism of neuronal degeneration leads attention to certain regions of the brain and connected throughout the brain that are impacted by Parkinson's disease. This work contributes to a more comprehensive knowledge of neural network disruptions in neurodegenerative disease and enhances the potential of GNN as diagnostic tool.

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Figures

References

Keywords

Back to Results